

Hackathon Project Phases Template that ensures students can complete it efficiently while covering all six phases. The template is structured to capture essential information without being time-consuming.

Hackathon Project Phases Template

Project Title:

LOGOCRAFT using Gen AI

Team Name:

Avengers

Team Members:

- Deepak Chilla
 - Achi vasu
 - Eshwar Dhanush Dhavaleswarapu
 - Boyina Venkata Sai Krishna
-

Phase-1: Brainstorming & Ideation

Objective:

- To develop an AI-powered logo creation tool using diffusion technology, enabling users to generate high-quality, customizable, and professional logos with minimal effort.

Key Points:

1. Problem Statement:

- In today's competitive market, businesses often encounter difficulties in crafting distinctive logos that authentically reflect their brand values.
- LogoCraft addresses this challenge by employing advanced Diffusion technology to generate bespoke logos tailored to user-provided descriptions.

2. Proposed Solution:

- Develop a AI-powered logo generator using diffusion technology with customizable prompts, inpainting tools, and SVG export, enabling users to create high-quality, professional logos effortlessly.

3. **Target Users:**

- Startups & Entrepreneur
- Freelancers & Designers
- Marketers and Content Creators
- Developers & Tech Enthusiasts
- Students & Hobbyists

4. **Expected Outcome:**

- A functional AI-powered logo design application that generates unique, professional-grade logos based on user-provided descriptions, utilizing advanced Diffusion technology.
-

Phase-2: Requirement Analysis

Objective:

- Define technical and functional requirements for Logo craft website

Key Points:

1. Technical Requirements:

- Languages: Python, JavaScript

Frameworks: PyTorch, Hugging Face Diffusers, OpenCV, NumPy fast-api uvicorn openai python-multipart

-

- Tools: Anaconda, VS Code, Google Colab, Inkscape, Photoshop

- Web Integration: HTML, CSS, FastAPI, Flask, Streamlit, React

- Workflow: Install model → Integrate with website → Generate logos → Refine & vectorize → Deploy

2. Functional Requirements :

- Users can upload text prompts describing the desired logo
- Users can specify design preferences (e.g., color scheme, typography, theme)
- The system utilizes a diffusion model to generate logos
- The system can refine the output based on additional user feedback.
- Users can refine generated logos using in-app tools (e.g., scaling, cropping, color).
- Users can regenerate parts of the logo without affecting the entire design (inpainting).
- Users can create accounts and save their generated logos.
- Role-based access control for different user types (e.g., free vs. premium users).
- An API is available for developers to use the logo-generation feature in other apks

3. Constraints & Challenges: (Any limitations or risks)

- **High Computational Requirements** – Requires powerful GPUs/TPUs for smooth performance.
- **Model Latency** – Generating high-quality logos takes time, impacting user experience.
- **Usability Challenges** – Non-technical users may find AI settings and customization difficult.
- **Scalability Issues** – Managing multiple concurrent users without performance lags is challenging.
- **Copyright and Legal Risks** – Risk of unintentionally generating logos similar to existing trademarks.

- **Integration Challenges** – Ensuring AI-generated logos work seamlessly with design tools like Adobe Illustrator and Figma.

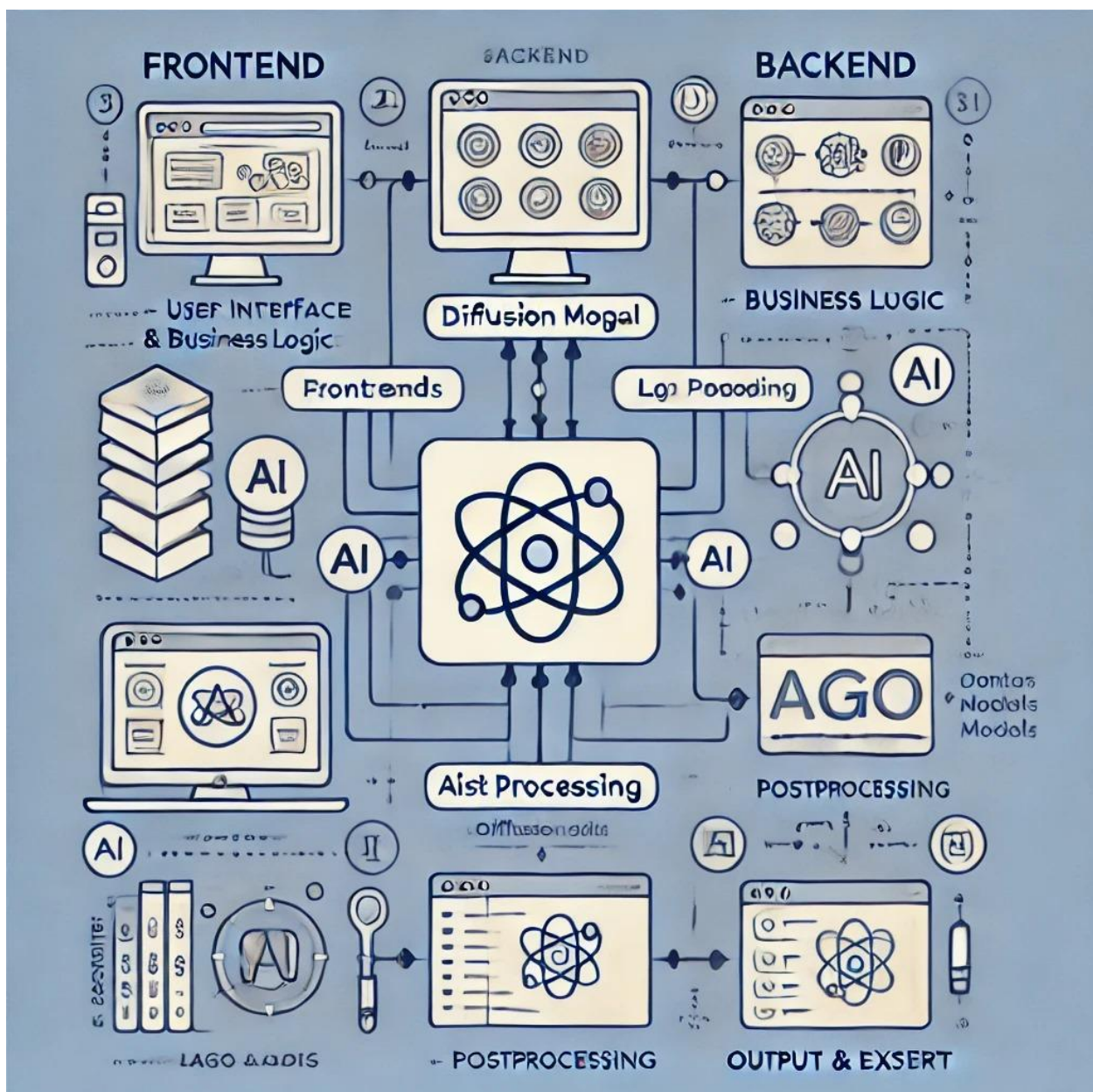
Phase-3: Project Design

Objective:

- Create the architecture and user flow.

Key Points:

1. **System Architecture Diagram:** (Simple sketch or flowchart)



- **User Interaction & Input:** The frontend allows users to input text prompts, upload images, and customize logo preferences.
- **AI Processing & Generation:** The backend processes inputs and utilizes a diffusion model to generate unique logo variations.
- **Export & Integration:** Users can refine logos, download them in various formats, and integrate them with design tools like Adobe Illustrator or Figma.

2. **User Flow:** (How a user will interact with the project)

1. **User Visits the Website**

- The user lands on the homepage and explores the features.
- Option to sign up/log in for saving projects.

2.**input & Customization**

- User enters a text prompt describing the logo.
- Option to upload reference images and select color themes, fonts, or styles.

3.**AI Processing & Logo Generation**

- The backend processes the input and sends it to the diffusion model.
- AI generates multiple logo variations.

4.**Preview & Customization**

- User reviews generated logos and selects the preferred one.
- Basic editing tools allow adjustments (colors, text, layout).

5.**Download & Export**

- User downloads the final logo in PNG, SVG, or JPEG formats.
- Option to integrate with external design tools like Adobe Illustrator or Figma.

• 6.**Save or Share**

- User can save the project for future edits or share it via a link.

3. **UI/UX Considerations:** (If applicable, wireframe or basic layout)

- **1.Simple & Intuitive Interface**

Clean, minimalistic design for ease of use.

Step-by-step guidance for new users.

- **2.User-Friendly Input Forms**

Clear input fields for text prompts, image uploads, and style preferences.

Live previews to show how inputs affect results.

- **3.Fast & Responsive AI Generation**

Real-time logo previews with smooth transitions.

Loading animations to indicate processing status.

- **4.Customizable Editing Tools**

Easy-to-use color pickers, font selectors, and layout adjustments.

Undo/redo functionality for smooth modifications.

- **5.Accessible & Mobile-Friendly Design**

Responsive UI optimized for desktops, tablets, and mobile devices.

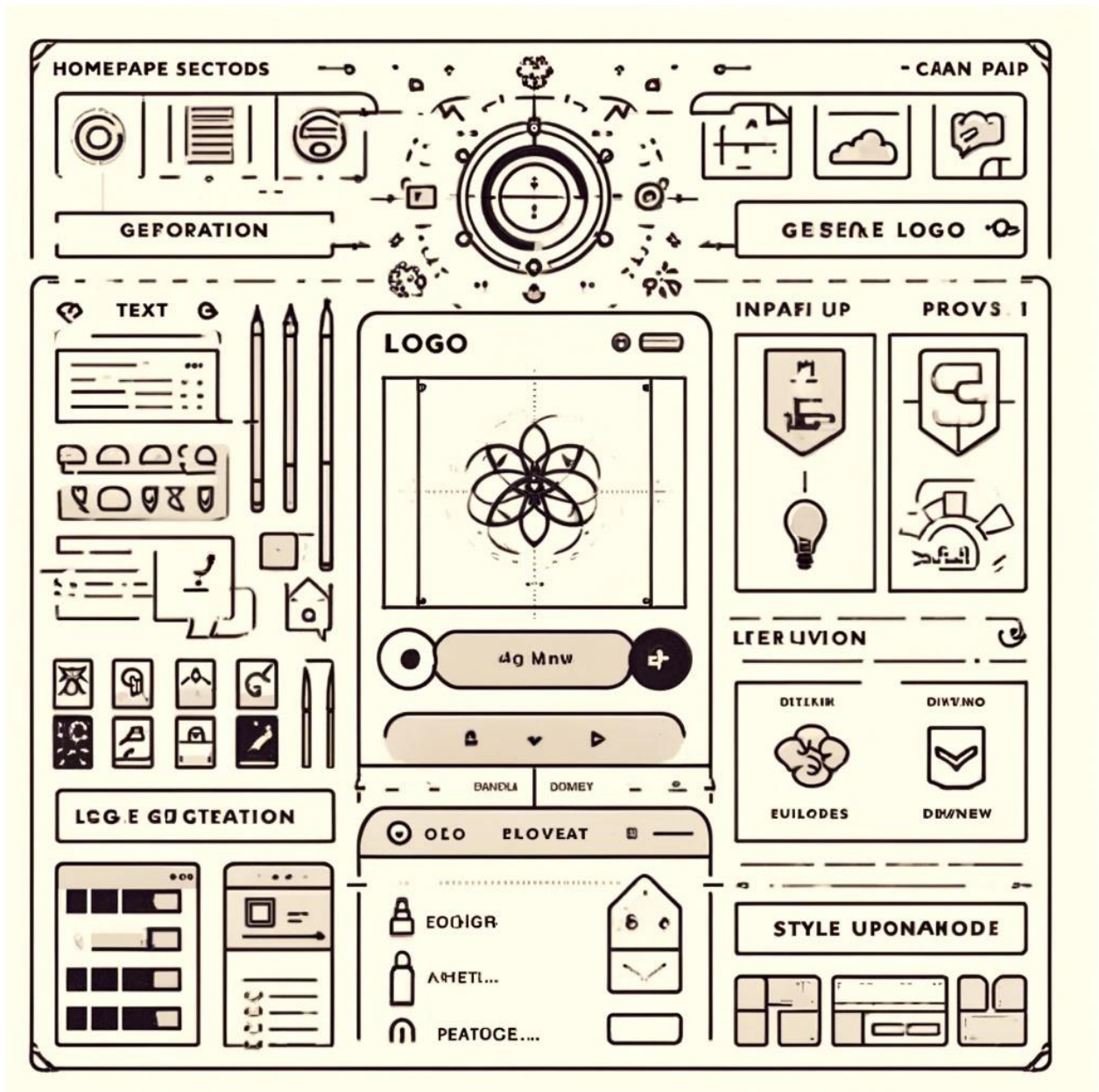
Dark mode and high-contrast options for better accessibility.

- **6.Download & Export Options**

Clear download buttons with multiple format choices (PNG, SVG, JPEG).

Integration with external design tools (Figma, Adobe Illustrator).

WIREFRAME OR BASIC LAYOUT WILL BE :



Phase-4: Project Planning (Agile Methodologies)

Objective:

- Break down the tasks using Agile methodologies in our website LOGOCRAFT

Key Points:

1. **Sprint Planning:** (Divide work into tasks for each team member)

Here’s the full table in a copyable format:

Sprint	Task	Priority	Duration	Deadline	Assigned To	Dependencies	Expected Outcome
Sprint 1	Research & Initial Setup	☐ High	1 Day	End of Day 1	Project Manager, UI/UX Designer, Backend Dev	Define scope, UI wireframe, setup server	Project plan & wireframes ready
Sprint 1	Frontend Basic UI Development	☐ Medium	6 hours (Day 1)	End of Day 1	Frontend Dev	Wireframes finalized	Basic UI structure completed
Sprint 2	AI Model Training & Integration	☐ High	1.5 Days (Day 2-3)	Mid-Day 3	AI/ML Engineer, Backend Dev	Dataset preparation, API setup	AI generating basic logos
Sprint 2	Backend API Development	☐ High	1 Day (Day 2)	End of Day 2	Backend Dev	AI Model integration	API handles user inputs & requests
Sprint 3	UI Enhancements & Customization	☐ Medium	6 hours (Day 3)	End of Day 3	Frontend Dev, UI/UX Designer	AI-generated logos available	Users can edit logos (colors, fonts)
Sprint 3	Final Testing & Deployment	☐ Low	1 Day (Day 4)	End of Day 4	Entire Team	Fully integrated system	Fully functional, demo-ready project

2. **Task Allocation:** (Who will do what?)

- **Project Manager:** Manages project timeline and team coordination.
- **UI/UX Designer:** Designs the website layout and user experience.
- **Frontend Developer:** Develops the user interface and customization tools.
- **Backend Developer:** Builds the API and manages database interactions.
- **AI/ML Engineer:** Trains and optimizes the AI-powered logo generation model.
- **QA Tester:** Tests the system for bugs and ensures smooth functionality.

3. **Timeline & Milestones:** (Set short deadlines for each task)

- **Day 1:** Project kickoff, research, and wireframe design.
- **Day 2:** Frontend UI development and backend setup.
AI model training and API integration.
UI enhancements, customization tools, and bug fixes.
- **Day 3:** Final testing, deployment, and project presentation.

Phase-5: Project Development

Objective:

- Code the project and integrate components for our LOGOCRAFT

Key Points:

1. **Technology Stack Used:** (List of programming languages, APIs, etc.)
 - Frontend: HTML,CSS,REACT JS
 - Backend: fast-api uvicorn openai python-multipart
 - Programming Language: Python,JS
2. **Development Process:** (Steps followed for coding)
 - **Setup & Planning:** Define requirements, choose tech stack, and create wireframes.
 - **Frontend Development:** Build UI components, user input fields, and customization tools.
 - **Backend Development:** Develop APIs, handle database storage, and connect AI model.
 - **AI Model Integration:** Train, fine-tune, and implement the diffusion model for logo generation.
 - **Testing & Debugging:** Identify and fix bugs, optimize performance, and ensure smooth functionality.
 - **Deployment & Final Review:** Deploy the project, conduct final testing, and prepare for launch.
3. **Challenges & Fixes:** (Mention any obstacles faced and how they were solved)
 - Challenge: Delayed API response times,it takes time to create complex designs
 - Fix: Implement caching to store frequently queried results.

Phase-6: Functional & Performance Testing

Objective:

- Ensure that LOGOCRAFT works as expected.
 - Functional & Performance Testing was putted in the box
4. Here's the table formatted for easy copying into MS Word:
 5.

Test Case ID	Category	Test Scenario	Expected Outcome	Status	Tester
TC-001	Functional Testing	User uploads an image for logo inspiration.	AI should generate a logo based on input.	✓ Passed	Tester 1
TC-002	Functional Testing	User customizes logo colors and fonts.	Customization should reflect in real time.	✓ Passed	Tester 2
TC-003	Performance Testing	AI logo generation time under 3 seconds.	AI should return results quickly.	❑ Needs Optimization	Tester 3
TC-004	Bug Fixes & Improvements	Fixed incorrect logo scaling issue.	Logos should maintain proper resolution.	✓ Fixed	Developer
TC-005	Final Validation	Ensure UI is responsive across devices.	UI should work on mobile & desktop.	✗ Failed - UI issues on mobile	Tester 2
TC-006	Deployment Testing	Host the app on a cloud-based platform.	The app should be accessible online.	❑ Deployed	DevOps

Final Submission

1. **Project Report Based on the templates**
 2. **Demo Video (3-5 Minutes)**
 3. **GitHub/Code Repository Link**
 4. **Presentation**
-